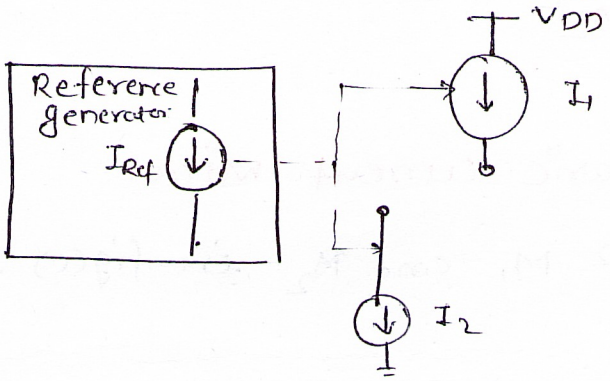
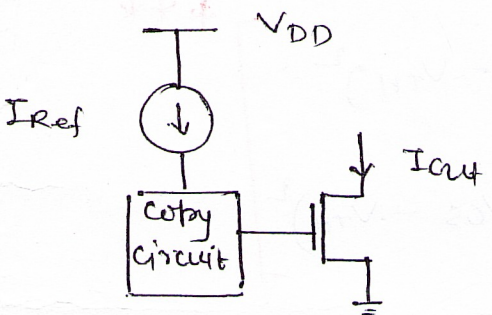
Chapter-5: Passive and Active Current Mirror.

Basic current mirror:-

→ The current mirror is an analog circuit is based on copying current from a reference, with assumption that one precisely defined current source is already available.



Conceptual means of copying current:-



fig(a)

How do we guarantee $I_{out} = I_{ref}$?

explanation:-

∵ for MOSFET, $I_D = f(V_{GS})$

⇒ $f(\cdot)$ denotes functionality of $\Rightarrow I_D$ vs V_{GS}

⇒ then, we can write, $V_{GS} = f^{-1}(I_D)$.

i.e. if transister is biased at I_{ref} then it produces

$V_{GS} = f^{-1}(I_{Ref})$ (as in fig(b))

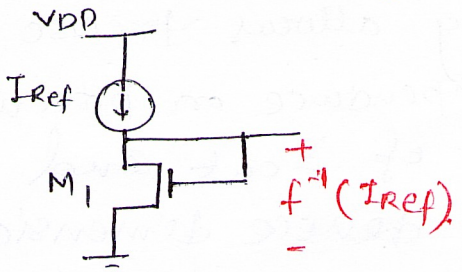


fig.(b)

now, if above V_{GS} is applied to the gate and source terminal of second MOSFET, the resulting current is,

$$I_{out} = f\left(f^{-1}(I_{Ref})\right) = I_{Ref}$$

(out in fig (c))

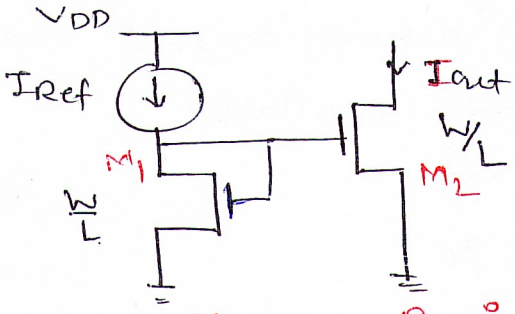


Fig. (c) → Basic current mirror.

→ Above structure of M_1 and M_2 in fig(c) is called a current mirror.

In general the device need not to be identical; Neglecting $\lambda = 0$, we can write,

$$\boxed{I_{Ref} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH})^2 \qquad I_{out} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{GS} - V_{TH})^2} \quad ***$$

$$\Rightarrow \boxed{I_{out} = \frac{(W/L)_2}{(W/L)_1} I_{Ref}}$$

for basic current mirror

Advantage:- Above topology allows precise copying of the current with no dependence on process and temperature. The ratio of I_{out} and I_{REF} is given by the ratio of device dimensions, a quantity that can be controlled with reasonable accuracy.

Cascode current mirror :-

→ In above basic current mirror, there are some non-ideal effect like channel length modulation does occure, especia if minimum length transistor are used.

→ To overcom from above problem we can go for cascode current ~~mirror~~ source as shown in below –

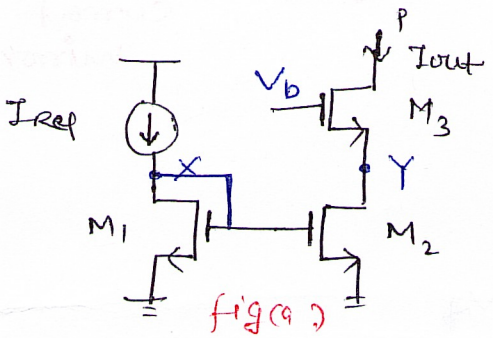


fig (a)

V_b choosen such that,

$$V_X = V_Y$$

then $I_{out} \approx I_{Ref}$.

Since the objective to ensure $V_Y = V_X$, we must guarante

$$V_b - V_{GS_3} = V_X \quad \Rightarrow \quad V_b = V_{GS_3} + V_X$$

⇒ if gate to source voltage is added to V_X the requir value of V_b can obtained as shown in fig (b)

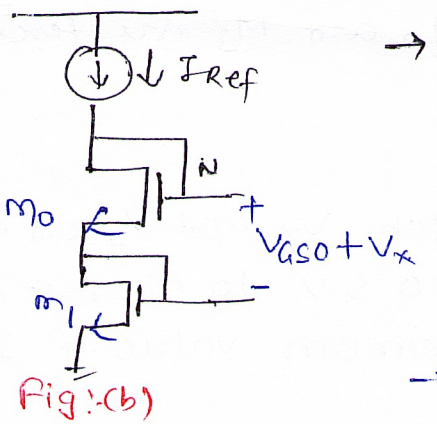


Fig :-(b)

→ Here, placing another diode connected device m_0 in series with m_1, thereby generating a voltage

$$V_N = V_{GS0} + V_X$$

→ There should be proper choice of m_0 with respect to m_2 yield,

$$V_{GS0} = V_{GS3}$$

Cascode current mirror circuit,

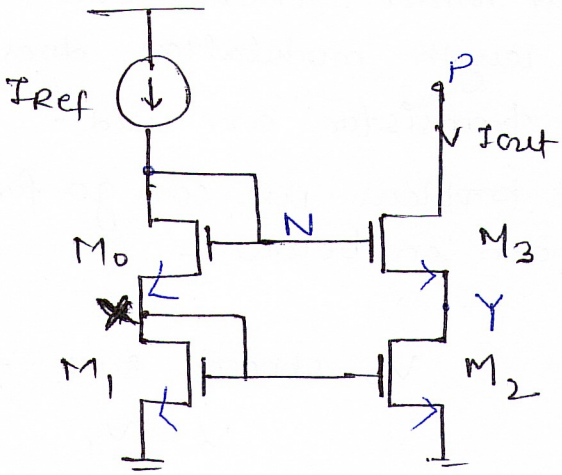


fig.(c) → cascode current mirror

connecting node N to gate of M_3,

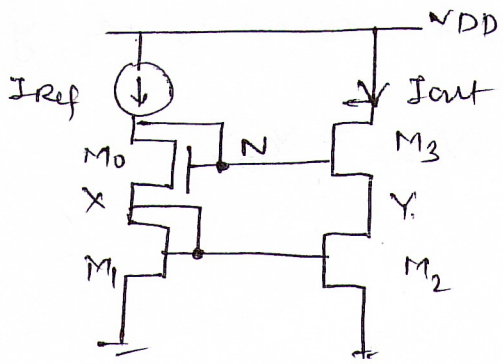
we have, $V_{GS0} + V_X = V_{GS3} + V_Y$

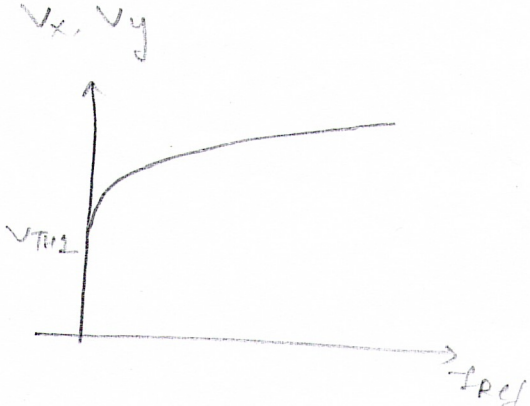
thus, if $$\frac{(W/L)_3}{(W/L)_0} = \frac{(W/L)_2}{(W/L)_1}$$

then, $V_{GS3} = V_{GS0}$ and $V_X = V_Y$

→ Above result hold even if M_0 and M_3 suffers from body effect.

Example: In below circuit sketch V_X and V_Y as a function of IRef. If Iref requires 0.5V to operate as a current source, what is maximum value of IRef.?





Solution:

∵ M_2 and M_3 are properly ratiod with respect to M_0 and M_1,

we have $V_x = V_y$

from ckt:- $V_x - V_{TH1} = \sqrt{\frac{2\ I_{Ref}}{\left[\mu_n C_{ox}\left(\frac{W}{L}\right)_1\right]}}$ $\left\{ I_{Ref} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_x - V_{t} \right.$

$\therefore V_x = \sqrt{\frac{2\ I_{Ref}}{\left[\mu_n C_{ox}\left(W/L\right)_1\right]}} + V_{TH1}$

now, to obtain maximum value of I_{Ref},

$V_N = V_{GS0} + V_{GS1}$

$= \left[\sqrt{\frac{2\ I_{Ref}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_0}} + V_{TH0}\right] + \left[\sqrt{\frac{2\ I_{Ref}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_1}} + V_{TH1}\right]$

$= \sqrt{\frac{2\ I_{Ref}}{\mu_n C_{ox}}}\left[\sqrt{\left(\frac{L}{W}\right)_0} + \sqrt{\left(\frac{L}{W}\right)_1}\right] + V_{TH0} + V_{TH1}$

from above circuit, we

$V_{DD} - (V_{GS0} + V_{GS1}) = 0.5$ (as given)

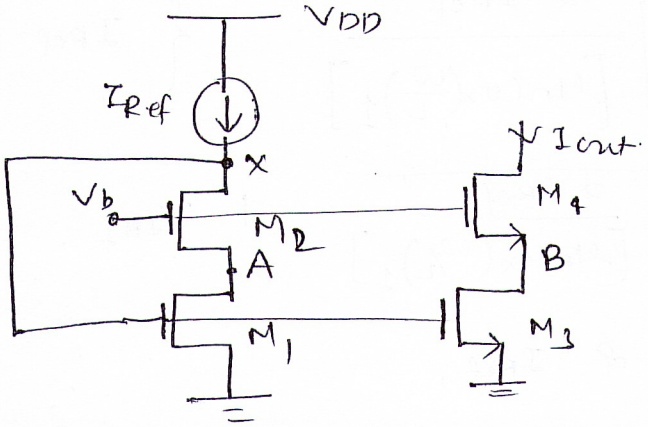
$\therefore V_{DD} - \left[\sqrt{\frac{2I_{Ref}}{\mu_n C_{ox}}}\left(\sqrt{\left(\frac{L}{W}\right)_0} + \sqrt{\left(\frac{L}{W}\right)_1}\right)\right] - V_{TH0} - V_{TH1} = 0.5$

rearranging above equation, we will get,

$$I_{Ref,max} = \frac{\mu_n C_{ox}}{2} \frac{\left(V_{DD} - V_{TH0} - V_{TH1} - 0.5\right)^2}{\left(\sqrt{\left(\frac{L}{W}\right)_0} + \sqrt{\left(\frac{L}{W}\right)_1}\right)^2}$$

Cascode mirror for Low-voltage operation :-

→ This circuit is in fact a cascode topology with its output shorted to its input.



Here, choose V_b so that M_1 and M_2 are in saturation.

then for this,

$$(V_b - V_{TH_2}) \leq V_x \ (= V_{GS1}) \quad -①$$

for M_2 to be saturated

~~for M_2 to~~

& $V_A (= V_b - V_{GS_2}) \geq V_{GS_1} - V_{TH_1}$ —②

for M_1 to be saturated.

i.e. from ① & ②

$$V_{GS_2} + (V_{GS_1} - V_{TH_1}) \leq V_b \leq (V_{GS_1} + V_{TH_1})$$

A solution exist if

$$V_{GS_2} + (V_{GS_1} - V_{TH_1}) \leq V_{GS_1} + V_{TH_2}$$

i.e. if $(V_{GS_2} - V_{TH_2}) < V_{TH_2}$.

we must therefore size M_2 such that its overdrive voltage remains less than one threshold voltage.

→ Now consider in above ckt, where all transister are in saturation and proper ratioing ensure that,

$$V_{GS_2} = V_{GS4}.$$

If $V_b = V_{GS2} + (V_{GS1} - V_{TH1}) = V_{GS4} + (V_{GS3} - V_{TH3})$

then cascode current source $M_3 - M_4$ consume minimum voltage while $M_1 - M_3$ sustain equal drain to source voltage, allowing copying of I_{Ref}. we call this low voltage cascode.

A

Active current Mirror:-

→ current mirrors can also process signals. i.e. operate as "active" element.

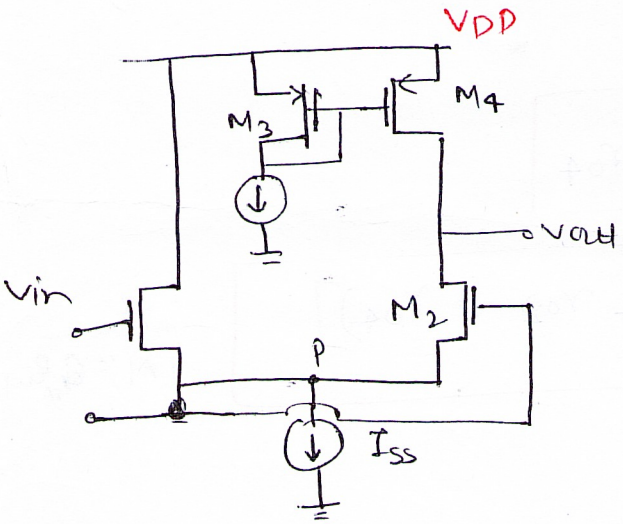


Fig:-(a) Differential pair with current-source load

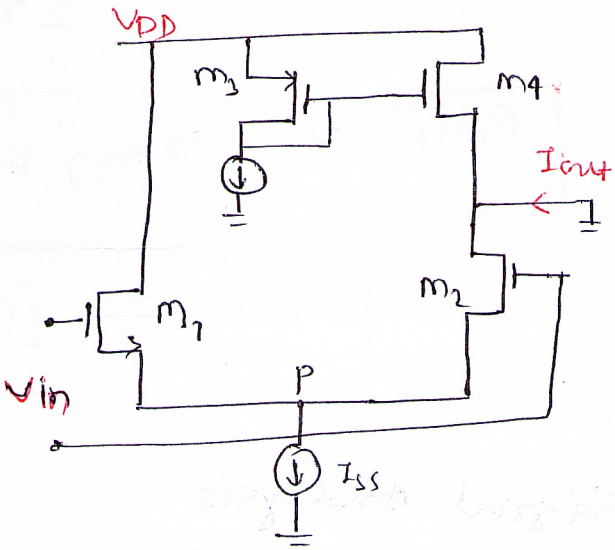


fig (b) Circuit for calculation of G_m

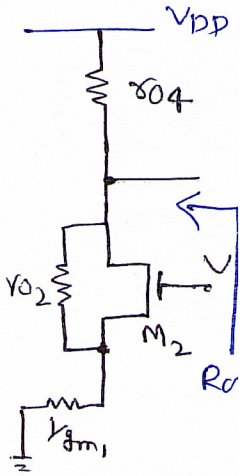


Fig (c) Circuit for calculation of R_{out}

Calculation of small signal gain = $|A_v| = \frac{V_{out}}{V_{in}}$

assume:- neglect effect (non-ideal effect)

we know, $|A_V| = G_m R_{out}$

from above fig. (b) $G_m = \frac{I_{out}}{V_{in}} = \frac{(g_{m_1} \cdot \frac{V_{in}}{2})}{V_{in}}$

$$\boxed{G_m = \frac{g_{m_1}}{2}}$$

calculation of R_{out} =?

In fig:(c), M_2 is degenerated by source o/p impedence

output impedence $= (1 + g_{m_2} r_{o_2}) \cdot \left(\frac{1}{g_{m_{1,2}}}\right) + r_{o_2}$

$= 2r_o + \frac{1}{g_{m_1}}$

$\simeq 2r_o$

$\therefore$ $\boxed{R_{out} = (2r_{o_2}) \parallel r_{o4}}$

$\therefore$ $\boxed{|A_V| = \frac{g_{m_1}}{2} [(2r_{o_2} \parallel r_{o4})]}$ $\because |A| = G_m R_{out}$

* Large-Signal Analysis
* Small-Signal Analysis
* Common-mode Properties.

*